Wenxiang Ying

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EDUCATION & APPOINTMENTS

University of Pennsylvania 2025 – present
- *Postdoctoral Researcher*
- *Advisor: Prof. Abraham Nitzan*

University of Rochester 2025
- *M.S. and Ph.D. in Chemistry*
- *Advisor: Prof. Pengfei (Frank) Huo*
- *Thesis: Theory and Quantum Dynamics Investigations of Polariton Chemistry, Photo-Physics, and Transport*

University of Science and Technology of China (USTC) 2020
- *B.S. in Chemical Physics*

SELECTED HONORS AND AWARDS

ACS Physical Chemistry Young Investigator Awards, ACS PHYS Division 2026
Chemical Computing Group Excellence Award for Graduate Students, ACS COMP Division 2025
Finalist of ACS Physical Chemistry Graduate Awards in Theoretical Chemistry, ACS PHYS Division 2024
Moses Passer Memorial Fellowship, University of Rochester 2024
Esther M. Conwell Graduate Fellowship, University of Rochester 2023
Excellent Teaching Assistant Award, USTC 2020, 2021

PUBLICATIONS

(∗ denotes the corresponding authors, # denotes authors contributed equally)

Preprint

[23]. **W. Ying***, M. Galperin*, and A. Nitzan*, "Vibrational Dressing of Collective Molecular Polaritons: A Scalable Green's-Function Theory", **arXiv:2610.20912** (2026);

Journal Articles

[22]. **W. Ying***, and A. Nitzan*, "A Mesoscopic Ginzburg–Landau Model for Vibrational Strong Coupling Enhanced Rayleigh Scattering in Molecular Liquids", **J. Phys. Chem. Lett.** 2026, 17 (39): 11186–11195;

[21]. **W. Ying***, and A. Nitzan*, "Excitation-Density-Controlled Regimes of Collective Light–Matter Dynamics", **J. Phys. Chem. Lett.** 2026, 17 (30): 8663–8669;

[20]. **W. Ying***, C. M. Bustamante, F. P. Bonafé, R. Richardson, M. Ruggenthaler, M. Sukharev*, A. Rubio*, and A. Nitzan*, "Linear and Nonlinear Vibrational Excitations Driven by Molecular Polaritons", **J. Chem. Theory Comput.** 2026, 22 (14): 7082–7095;

[19]. C. M. Bustamante*, F. P. Bonafé, R. Richardson, M. Ruggenthaler, **W. Ying**, A. Nitzan, M. Sukharev, and A. Rubio, "Collective Rabi-Driven Vibrational Activation in Molecular Polaritons", **Nano Lett.** 2026, 26 (15): 5298–5306;

[18]. **W. Ying***, and A. Nitzan*, "Electron transfer in confined electromagnetic fields: A unified Fermi's golden rule rate theory, and extension to lossy cavities", **J. Chem. Phys.** 2026, 164 (2): 024113;

[17]. **W. Ying**, M. E. Mondal, E. R. Koessler, S. M. Vega, and P. Huo*, "Collective Effects in Polariton Chemistry and Photophysics", **Annu. Rev. Phys. Chem.** 2026, 77: 201-224;

[16]. Y. Lai*, **W. Ying**, T. D. Krauss, and P. Huo*, "Analytic Rate Theory of Polariton Relaxation that Explains Long Polariton Lifetime", **J. Chem. Phys.** 2026, 164 (2): 024103;

[15]. **W. Ying**[#], B. X.-K. Chng[#], M. Delor, and P. Huo*, "Microscopic Theory of Polariton Group Velocity Renormalization", **Nat. Commun.** 2025, 16 (1): 6950;

[14]. S. M. Vega[#], **W. Ying**[#], and P. Huo*, "Theoretical Insights into the Resonant Suppression Effect in Vibrational Polariton Chemistry", **J. Am. Chem. Soc.** 2025, 147 (23): 19727–19737;

[13]. D. Hu*, B. X.-K. Chng, **W. Ying**, and P. Huo*, "Trajectory-based Non-adiabatic Simulations of the Polariton Relaxation Dynamics", **J. Chem. Phys.** 2025, 162 (12): 124113;

[12]. B. X.-K. Chng, M. E. Mondal, **W. Ying**, and P. Huo*, "Quantum Dynamics Simulations of Exciton Polariton Transport", **Nano Lett.** 2025, 25 (4): 1617-1622;

[11]. B. X.-K. Chng, **W. Ying**, Y. Lai, A.N. Vamivakas, S.T. Cundiff, T.D. Krauss*, and P. Huo*, "Mechanism of Polariton Decoherence in the Collective Light-Matter Couplings Regime", **J. Phys. Chem. Lett.** 2024, 15 (47): 11773–11783;

[10]. **W. Ying***, Y. Su, Z.-H. Chen, Y. Wang*, and P. Huo*, "Spin Relaxation Dynamics with a Continuous Spin Environment: the Dissipaton Equation of Motion Approach", **J. Chem. Phys.** 2024, 161 (14): 144112;

[9]. Y. Lai*, **W. Ying**, and P. Huo*, "Rate Theory and Quantum Master Equation for Polariton Relaxation Dynamics", **J. Chem. Phys.** 2024, 161 (10): 104109;

[8]. **W. Ying***, M. E. Mondal, and P. Huo*, "Theory and Quantum Dynamics Simulations of Exciton-Polariton Motional Narrowing", **J. Chem. Phys.** 2024, 161 (6): 064105;

[7]. **W. Ying**, and P. Huo*, "Resonance Theory of Vibrational Strong Coupling Enhanced Polariton Chemistry and The Role of Photonic Mode Lifetime", **Commun. Mater.** 2024, 5 (1): 110;

[6]. **W. Ying**, M. A.D. Taylor, and P. Huo*, "Resonance Theory of Vibrational Polariton Chemistry at the Normal Incidence", **Nanophotonics** 2024, 13 (14): 2601-2615;

[5]. D. Hu, **W. Ying***, and P. Huo*, "Resonance Enhancement of Vibrational Polariton Chemistry Obtained from the Mixed Quantum-Classical Dynamics Simulations", **J. Phys. Chem. Lett.** 2023, 14 (49): 11208–11216;

[4]. **W. Ying***, and P. Huo*, "Resonance Theory and Quantum Dynamics Simulations of Vibrational Polariton Chemistry", **J. Chem. Phys.** 2023, 159 (8): 084104;

[3]. D. Bossion*, **W. Ying**, S.N. Chowdhury, and P. Huo*, "Non-adiabatic Mapping Dynamics in the Phase Space of $SU(N)$ Lie Group", **J. Chem. Phys.** 2022, 157 (8): 084105;

[2]. S. Chen, B.B. Gong, J. Gu, Y. Lin, B. Yang, Q.Q. Gu, R. Jin, Q. Liu, **W. Ying**, X.X. Shi, W.L. Xu, L.H. Cai, Y. Li, Z.H. Sun, S.Q. Wei, W.H. Zhang, and J.L. Lu, "Dehydrogenation of Ammonia Borane by Platinum–Nickel Dimers: Regulation of the Heteroatom Interspace Boosts Bifunctional Synergetic Catalysis," **Angew. Chem. Int. Ed.** 2022, 61 (41): e202211919;

[1]. S. Chen, L. Huang, Z.H. Sun, L.N. Cao, **W. Ying**, X.X. Shi, W. Liu, J. Gu, X.S. Zheng, J.F. Zhu, Y. Lin, S.Q. Wei, and J.L. Lu, "Synthesis of Quasi-Bilayer Subnano Metal-Oxide Interfacial Cluster Catalysts for Advanced Catalysis," **Small** 2020, 16 (52): 2005571;

TEACHING

- **University of Pennsylvania**: Teaching Assistant for Chem 5260 Chemical Dynamics (Spring 2026).
- **University of Rochester**: Teaching Assistant for CHM 469 & 470 Computational Chemistry I & II (Spring 2023, Spring 2024), CHM 132 General Chemistry Lab (Spring 2023), CHM 252/442 Physical Chemistry II (Spring 2022), CHM 251/441 Physical Chemistry I (Fall 2021).
- **USTC**: Teaching Assistant for 003158(.03/.05/e.01/.01) Quantum Physics (Fall 2019, Spring 2020, Fall 2020, Spring 2021).

MENTORSHIP & SERVICE

- **Graduate Students**: Benjamin X.-K. Chng, Sebastian M. Vega, Santanu Poddar, Rahul Chourasiya.
- **Undergraduate Students**: Mateo L. Castellanos (currently PhD student at Stony Brook University).
- **Contributing Reviews to Scientific Journals**: *Science*, *Scientific Reports*, *Journal of the American Chemical Society*, *The Journal of Physical Chemistry Letters*, *Journal of Chemical Theory and Computation*, *The Journal of Chemical Physics*.

OTHER RESEARCH EXPERIENCE, WORKSHOPS & SYMPOSIA

California Institute of Technology, Pasadena, CA, United States 11.30.2026 - 12.03.2026
- *Marcus Center Winter School on Quantum Many-Body Simulation*
- *Lecturers: Garnet Chan, Xing Zhang, Huanchen Zhai, Johnnie Gray, Tianyu Zhu, Sandeep Sharma, Lin Lin*

Max-Planck Institute for the Structure and Dynamics of Matter, Hamburg, Germany 04.20.2026 - 04.22.2026
- *Rising Stars Workshop*
- *Organizers: Philip Moll, Angel Rubio, Andrea Cavalleri, Jie Shan, Kin-Fai Mak*

Max-Planck Institute for the Structure and Dynamics of Matter, Hamburg, Germany 10.26.2025 - 10.31.2025
- *Unravelling the Mysteries of Vibrational Strong Coupling (UnMySt) Theory Workshop / Hackathon*
- *Organizers: Dominik Sidler, Angel Rubio*

Flatiron Institute, New York, NY, United States 09.29.2025 - 10.03.2025
- *Ab Initio Quantum Electrodynamics for Quantum Materials Engineering School*
- *Organizers: Simone Latini, Johannes Flick, Michael Ruggenthaler, Angel Rubio*

Arizona State University, Phoenix, AZ, United States 09.09.2025 - 09.28.2025
- *Conducting finite-difference time-domain (FDTD) simulations of molecular ensembles in confined electromagnetic environments.*
- *Host: Prof. Maxim Sukharev*

Institute of Software, Chinese Academy of Sciences, Beijing, China 06.15.2023 - 07.14.2023
- *Developing quantum algorithms for chemical dynamics simulations.*
- *Hosts: Prof. Shenggang Ying, Prof. Mingsheng Ying*

USTC, Hefei, Anhui, China 09.01.2020 - 07.15.2021
- *Exploring open quantum dynamics with non-Gaussian environments.*
- *Hosts: Prof. Xiao Zheng, Prof. YiJing Yan*

Boston University, Boston, MA, United States 07.05.2019 - 08.31.2019
- *Developing nonadiabatic molecular dynamics methods based on the partially linearized density matrix (PLDM) approach.*
- *Host: Prof. David F. Coker*

INVITED TALKS AT UNIVERSITIES, INSTITUTIONS & LABORATORIES

11. 04/21/2026, Rising Stars Workshop, Max-Planck Institute for the Structure and Dynamics of Matter, Hamburg, Germany.
10. 10/26/2025, Max-Planck Institute for the Structure and Dynamics of Matter, Hamburg, Germany.
9. 09/04/2025, UnMySt Virtual Seminar, European Research Council (ERC) Synergy Project.
8. 06/28/2025, Beijing Normal University, Zhuhai, Guangdong, China.
7. 06/27/2025, South China Normal University, Guangzhou, Guangdong, China.
6. 06/24/2025, Westlake University, Hangzhou, Zhejiang, China.
5. 06/23/2025, USTC, Hefei, Anhui, China.
4. 02/17/2025, Virtual Talk at Professor Abraham Nitzan's Group, University of Pennsylvania.
3. 10/22/2024, Virtual Talk at Professor Joonho Lee's Group, Harvard University.
2. 09/25/2024, ACS PHYS Graduate Award Virtual Symposium.
1. 07/26/2023, Hefei National Research Center for Physical Sciences at the Microscale & USTC, Hefei, Anhui, China.

CONTRIBUTIONS TO INTERNATIONAL CONFERENCES

9. W. Ying, C. M. Bustamante, F. P. Bonafé, R. Richardson, M. Ruggenthaler, M. Sukharev, A. Rubio, and A. Nitzan, "Linear and Nonlinear Vibrational Activation Driven by Molecular Polaritons", oral, at ACS Fall Meeting, Chicago, IL, United States, August 23-27, 2026.
8. W. Ying, C. M. Bustamante, F. P. Bonafé, R. Richardson, M. Ruggenthaler, M. Sukharev, A. Rubio, and A. Nitzan, "Linear and Nonlinear Vibrational Activation Driven by Molecular Polaritons", poster, at Molecular Interactions and Dynamics Gordon Research Conference, North Easton, MA, United States, Jun 28-Jul 03, 2026.

7. W. Ying, and P. Huo, "Quantum Dynamics Simulations of Polariton Chemistry, Photo Physics, and Transport", poster, at ACS Fall Meeting, Washington, DC, United States, August 17-21, 2025.

6. W. Ying, and P. Huo, "Quantum Dynamics Simulations of Polariton Chemistry, Photo Physics, and Transport", poster, at Penn Conference on Theoretical Chemistry (PCTC), Philadelphia, PA, United States, May 2-4, 2025.

5. W. Ying, M. E. Mondal, and P. Huo, "Theory and Quantum Dynamics Simulations of Exciton-Polariton Motional Narrowing", poster, at ACS Spring Meeting, San Diego, CA, United States, March 23-27, 2025.

4. W. Ying, M. E. Mondal, and P. Huo, "Theory and Quantum Dynamics Simulations of Exciton-Polariton Motional Narrowing", oral and poster, at APS March Meeting, Anaheim, CA, United States, March 16–21, 2025.

3. W. Ying, M. E. Mondal, and P. Huo, "Theory and Quantum Dynamics Simulations of Exciton-Polariton Motional Narrowing", poster, at American Conference on Theoretical Chemistry (ACTC), Chapel Hill, NC, United States, June 17-20, 2024.

2. W. Ying and P. Huo, "Resonance Theory of Vibrational Polariton Chemistry", oral and poster, at APS March Meeting, Minneapolis, MN, United States, March 3-8, 2024.

1. W. Ying and P. Huo, "Chemical Reactivity Under Vibrational Strong Coupling", poster, at Quantum Control of Light and Matter Gordon Research Conference, Newport, RI, United States, August 6-11, 2023.

PROFESSIONAL REFERENCES

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Prof. Angel Rubio
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Co-Director, Initiative for Computational Catalysis
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